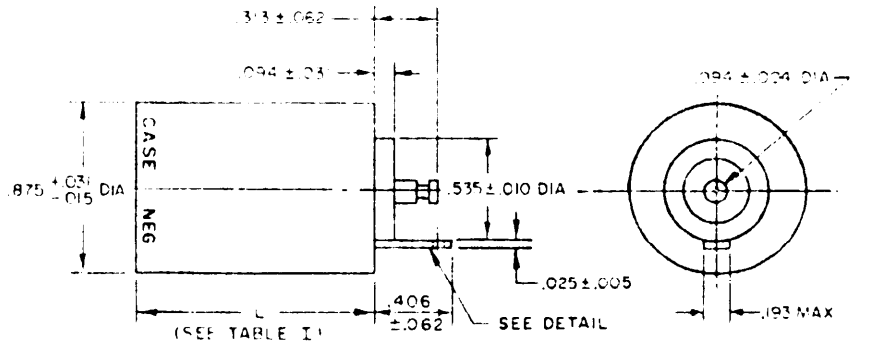
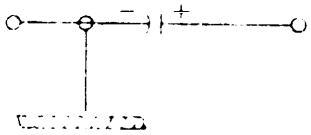
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR14

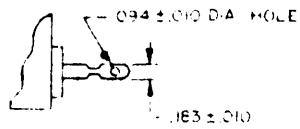
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39006.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





CIRCUIT DIAGRAM



DETAIL OF SOLDER LUG TERMINAL

INCHES	MM	INCHES	MM
.004	.10	.094	2.39
.005	.13	.183	4.65
.010	.25	.193	4.90
.015	.38	.313	7.95
.025	.64	.406	10.31
.031	.79	.535	13.59
.062	1.57	.875	22.23

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CLR14 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Rated voltage (85 °C)	Cap.	DC leakage (max)			Dissipation factor (max)	Derated voltage (125 °C)	Surge voltage (85 °C)	Impedance (max)	Dimension L ±.062(1.57)	Part No. M39006/19- Failure rate level (% 1,000 hr)				
		25 °C	85 °C	125 °C						L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	µF	µA	µA	µA	percent	volts, dc	volts, dc	ohms	inches					
20	100.0	10	50	80	21.0	18	23	30	0.540 (13.72)	1000	1020	1040	1060	1080
20	200.0	16	80	128	36.0	18	23	20	0.732 (18.59)	1001	1021	1041	1061	1081
30	75.0	11	55	88	15.0	26	34.5	45	0.540 (13.72)	1002	1022	1042	1062	1082
30	150.0	13	90	104	29.0	26	34.5	30	0.732 (18.59)	1003	1023	1043	1063	1083
60	40.0	12	60	96	8.2	52	69	65	0.540 (13.72)	1004	1024	1044	1064	1084
60	80.0	19	95	152	16.0	52	69	35	0.732 (18.59)	1005	1025	1045	1065	1085
90	25.0	11	55	88	5.1	78	103	90	0.540 (13.72)	1006	1026	1046	1066	1086
90	50.0	18	90	144	10.0	78	103	45	0.732 (18.59)	1007	1027	1047	1067	1087
180	12.0	11	55	88	5.1	155	207	180	0.920 (23.37)	1008	1028	1048	1068	1088
180	25.0	18	90	144	10.0	155	207	90	1.300 (33.02)	1009	1029	1049	1069	1089
270	8.0	11	55	88	5.1	235	310	270	1.270 (32.36)	1010	1030	1050	1070	1090
270	16.0	18	90	144	10.0	235	310	135	1.865 (47.37)	1011	1031	1051	1071	1091
360	6.0	11	55	88	5.0	310	414	360	1.635 (41.53)	1012	1032	1052	1072	1092
360	12.0	18	90	144	10.0	310	414	180	2.420 (61.47)	1013	1033	1053	1073	1093
450	5.0	11	55	88	4.9	390	518	450	2.000 (50.80)	1014	1034	1054	1074	1094
450	10.0	18	90	144	9.8	390	518	225	2.980 (75.69)	1015	1035	1055	1075	1095
540	4.0	11	55	88	5.1	470	621	540	2.365 (60.07)	1016	1036	1056	1076	1096
540	8.0	18	90	144	10.0	470	621	270	3.532 (89.71)	1017	1037	1057	1077	1097
630	3.5	11	55	88	5.0	545	724	630	2.720 (69.09)	1018	1038	1058	1078	1098
630	7.0	18	90	144	10.0	545	724	315	4.062 (103.17)	1019	1039	1059	1079	1099

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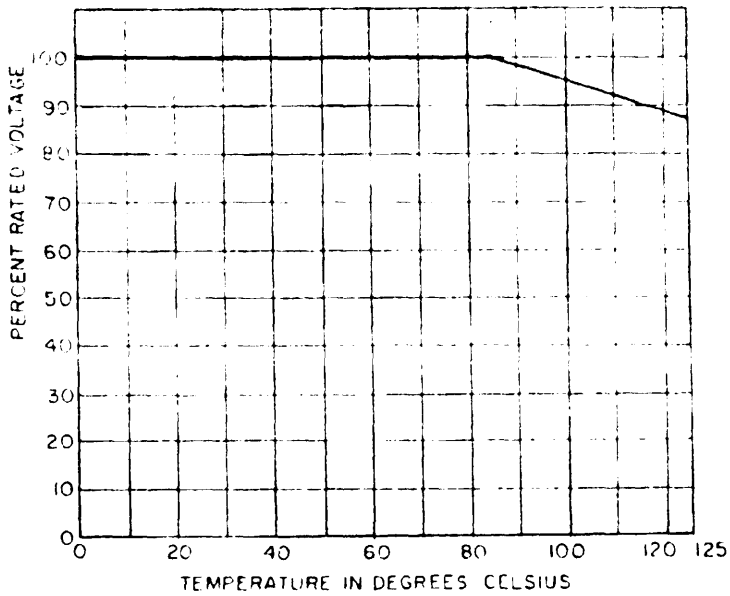


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, uninsulated.
Seal type - G (hermetic).
Terminals - Solder-lug.
Rated temperature - -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - +75, -15%.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Not applicable.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Bend (if required) - Method 211 of MIL-STD-202, condition B.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±8% of initial measurement.
DF - 115% (max) of initial requirement.

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Dielectric withstanding voltage (DWV):
Not applicable.

Insulation resistance (IR): Not applicable.

Low temperature (storage): Method 502 of MIL-STD-810.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL - See table I.
Cap. - Within tolerance of table I value.
DF - See table I.
Step 2 (-55°C):
Imp. - See table I.
ΔCap. - -60% (20v), -45% (30v), or -35% (>30v) of step 1 value (max).
Step 3 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.
Step 4 (+85°C):
DCL - See table I.
ΔCap. - +15% (≤30v) or +10% (>30v) of step 1 value (max).
DF - See table I.
1/ Step 5 (+125°C):
DCL - See table I.
ΔCap. - +20% (≤26v) or +10% (>26v) of step 1 value (max).
DF - See table I.
Step 6 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.
2,000-hr (qualification):
DCL - See table I.
ΔCap. - Within ±15% of initial measurement.
DF - See table I.
240-hr (group B):
DCL - See table I.
ΔCap. - Within ±10% of initial measurement.
DF - See table I.

10,000-hr (group C) and extended life:
DCL (85°C) - 125% (max) of table I value.
DCL (25°C) - See table I.
ΔCap. - Within ±25% of initial measurement.
DF - 200% (max) of table I value.
DWV and IR - Not applicable.

Vacuum life: Not applicable.

Life at 125°C: Method 108 of MIL-STD-202.
DCL - See table I.
ΔCap. - Within ±15% of initial measurement.
DF - 130% (max) of table I value.
DWV and IR - Not applicable.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
Test voltage - Rated voltage for 20 to 450v units; 500 vdc for 540 and 630v units.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Marking: Capacitors shall be marked as shown in the following example:

EXAMPLE:

M39006/19-1000	- Part number.
JAN (TM)	- "JAN" brand and trademark.
12345 7015A	- Source code, date code, and lot symbol.
100μF 20VDC	- Capacitance and rated voltage.

Polarity - Capacitors shall be marked "CASE NEG" on the negative end (see figure 1).

Voltage groups (85°C rated voltage):

I	20 to 90v incl
II	180 to 360v incl
III	450 to 630v incl

1/ 125°C derated voltage referenced.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R001

INSTRUCTIONS

This sheet is to be filled out by personnel either (government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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